Functional Requirements Document (FRD)

Production-Grade Multi-Tenant AI Recruitment ATS — Specification & Scope Baseline

DOCUMENT ID	SYSTEM ARCHITECTURE	MULTI-TENANCY	EFFECTIVE DATE
FRD-ATS-2026-V1	7-Layer FastAPI + Next.js 15	Hard Isolated (org_id)	October 2026

1. EXECUTIVE SUMMARY & CORE PLATFORM SCOPE

The Multi-Tenant AI Recruitment Applicant Tracking System (ATS) is an enterprise SaaS backend and client platform designed for scalable talent acquisition. The platform enforces strict organization data isolation (`organization_id`), $0-cost keyword full-text search (`tsvector`), evidence-backed candidate-to-job matching, candidate pipeline state workflows, and RFC-5545 compliant asynchronous calendar scheduling.

2. 7-LAYER ARCHITECTURAL DATA FLOW



3. SYSTEM USER ROLES & ACCESS CONTROL HIERARCHY

Role	Tenant Scope	Permitted Actions & System Boundaries
System Admin	Global Cross-Tenant	Provision new organizations, system maintenance, audit log inspection.
Tenant Owner / Admin	Organization Bound	Manage recruiters, configure job boards, view company analytics, manage integrations.
Recruiter	Organization Bound	Post jobs, parse resumes, run candidate searches, advance application stages, book interviews.
Hiring Manager	Organization Bound	Review matched candidates, complete interview scorecards, approve stage movements.
Interviewer	Assigned Interviews	View assigned candidate profiles, join meeting links, submit evaluation rubrics.

4. FUNCTIONAL REQUIREMENTS: CORE ENGINE (MILESTONES 1 – 10 BASELINE)

Req ID	Module	Functional Requirement Specification	Status
FR-101	Authentication	User registration and login issuing signed JWT tokens containing `user_id` and `organization_id` claims.	COMPLETED
FR-102	Multi-Tenancy	Strict multi-tenant row-level filtering (`WHERE organization_id = :org_id`) across all SQL queries and endpoints.	COMPLETED
FR-201	Job Management	CRUD operations for job postings, activation/deactivation state management, and organizational scoping.	COMPLETED
FR-301	Candidate Hub	Candidate profile creation, resume uploading with file size/MIME validation, and private disk/S3 storage keys.	COMPLETED
FR-302	CV Extraction	Automated text extraction from PDF and DOCX resume uploads, populating database `raw_text` attributes.	COMPLETED
FR-303	Keyword Search	Instant $0-cost full-text keyword search across candidate resumes using PostgreSQL `to_tsvector` & `to_tsquery`.	COMPLETED
FR-401	Job Matching	Evidence-backed deterministic scoring engine evaluating candidate `raw_text` against weighted job criteria.	COMPLETED
FR-501	Pipeline Workflow	Application state machine supporting status transitions (`applied`, `screening`, `interview`, `offer`, `hired`, `rejected`).	COMPLETED
FR-502	Interview Core	Interview scheduling linking candidates, applications, recruiters, and internal/external interviewers.	COMPLETED
FR-503	Calendar (.ics)	RFC-5545 compliant `.ics` payload generator dynamically binding the authenticated recruiter email as `ORGANIZER`.	COMPLETED
FR-504	Async Dispatch	FastAPI `BackgroundTasks` non-blocking email dispatch delivering `.ics` calendar invites and meeting links.	COMPLETED

> **✓ Verified Backend Quality Milestone Milestone Verification**
> Milestones 1 through 10 are completely implemented in the backend codebase, backed by 87 passing black-box integration tests (`pytest`). All domain services raise custom exceptions mapped at the FastAPI route layer, eliminating HTTP framework leakage.

5. FUNCTIONAL REQUIREMENTS: CLIENT UI, AI & RAG (MILESTONES 11 – 13)

Req ID	Module	Functional Requirement Specification	Status
FR-601	Next.js Client	Next.js 15 App Router dashboard with JWT route protection and tenant context management.	PENDING - M11
FR-602	Kanban Board	Interactive drag-and-drop application pipeline UI with optimistic state updates and rollback safety.	PENDING - M11
FR-603	Resume Viewer	Split-screen profile interface with embedded PDF resume previewer and highlighted match keywords.	PENDING - M11
FR-701	Vector Indexing	PostgreSQL `pgvector` extension integration storing CV chunk embeddings with HNSW index acceleration.	PLANNED - M12
FR-702	Hybrid Retrieval	Reciprocal Rank Fusion (RRF) algorithm combining `tsvector` keyword search with dense vector similarity.	PLANNED - M12
FR-801	Gemini AI RAG	Evidence-grounded candidate evaluation generating structured JSON match scores via Gemini API.	PLANNED - M13
FR-802	Citation Guard	Strict citation verification requiring LLM outputs to reference exact verbatim quotes from candidate CVs.	PLANNED - M13

6. SEARCH ARCHITECTURE TRANSITION: KEYWORD VS. HYBRID VECTOR RAG

Phase 1: Full-Text Search (M7) CURRENT

- **Engine:** PostgreSQL `tsvector` & `tsquery`.
- **Cost:** $0 infrastructure overhead.
- **Strength:** Instant exact match for specific technical terms (e.g., "FastAPI", "Docker").
- **Limitation:** Misses semantic synonyms (e.g., searching "AWS" won't match "Amazon Web Services").

Phase 2: Hybrid RAG Search (M12-M13) TARGET

- **Engine:** `tsvector` + `pgvector` + RRF Fusion.
- **Intelligence:** Gemini 2.5 structured RAG scoring.
- **Strength:** Understands conceptual experience, skill transferability, and context.
- **Guardrail:** Mandatory citation quotes eliminate AI hallucinations.

7. FUNCTIONAL REQUIREMENTS: COLLABORATION, SCALE & OPS (MILESTONES 14 – 16)

Req ID	Module	Functional Requirement Specification	Status
FR-901	Scorecards	Configurable evaluation rubrics with weighted scoring (1-5) and hire/no-hire recommendations.	PLANNED - M14
FR-902	Notes & Feed	Collaborative activity timeline supporting team `@mentions` and field-level compensation privacy.	PLANNED - M14
FR-1001	Celery Queue	Distributed background task queue backed by Redis for heavy PDF parsing and LLM inference processing.	PLANNED - M15
FR-1002	Observability	Structured JSON logging with unique `X-Request-ID` tracing and LLM token usage cost tracking.	PLANNED - M15
FR-1101	Calendar OAuth	2-way live Google Calendar & Microsoft Graph OAuth sync generating dynamic Meet/Teams links.	PLANNED - M16
FR-1102	Analytics	Executive recruitment funnel analytics tracking Time-to-Hire, conversion rates, and sourcing ROI.	PLANNED - M16

8. MILESTONE GAP ANALYSIS & COMPONENT TRACEABILITY MATRIX

Milestone	Backend Component	Frontend Component	Gap Status
M1 – M10	API Routers, Services, DB Models, `.ics` Gen	None (Postman / Pytest Tested)	BACKEND READY
M11 (Next)	API Client Cors / Middleware Updates	Next.js 15 Dashboard & Kanban Board UI	FRONTEND MISSING
M12	`pgvector` Extension & HNSW Indexing	Semantic Search UI & Keyword Toggle	PENDING DEV
M13	Gemini RAG Service & Citation Engine	AI Match Rationale & Citation Badge UI	PENDING DEV
M14 – M16	Celery Workers, OAuth Sync, Analytics Views	Scorecard Modal, Analytics Charts UI	PENDING DEV

9. NON-FUNCTIONAL REQUIREMENTS (NFRS) & QUALITY BENCHMARKS

Category	Benchmark & System Requirement Specification
Performance	API endpoint p95 response time < 150ms for non-AI queries; search execution < 100ms across 100k records.
Security	Bcrypt password hashing (work factor 12), JWT token expiration, zero cross-tenant data leak tolerance.
Availability	99.9% system uptime; stateless FastAPI application nodes allowing horizontal scaling behind load balancers.
Data Privacy	Private S3 / local disk storage for candidate resumes; secure signed URL access controls for document downloads.
Maintainability	Strict 7-layer architecture; 100% type annotations in Python and TypeScript; black-box API integration tests.

10. ENTERPRISE COMPLIANCE & SECURITY GUARDRAILS

Tenant Data Isolation

All database queries enforce `organization_id` parameters at the repository layer. Cross-tenant request attempts automatically trigger custom `ForbiddenError` exceptions.

Prompt Injection Defense

Extracted CV raw text is sanitized prior to LLM RAG processing, neutralizing embedded prompt injection instructions hidden inside resume documents.

11. DOCUMENT REVIEW & ARCHITECTURAL SIGN-OFF

Lead Systems Architect

Status: APPROVED

Lead Backend Engineer

Status: VERIFIED (M1-M10)

Product Owner

Status: ACCEPTED